

CHAPTER 1

INTRODUCTION

1.1 ABOUT

About 7 million people living along the Indian coastline, spanning over 8100 km, are depending on fishing for their livelihood. Locating and catching fish is always a challenging task. Often, the search for fish ends up in spending considerable time and resources, thus increasing the cost leading to low profitability. A reliable and timely advisory on the potential zones of fish aggregation will benefit the fishing community to reduce the time and effort spent in searching the shoals of fish, thus improving the profitability and hence, the socio-economic status.

1.2 FISHERY STATISTICS

The fishery statistics provides the nations perspective in the development of fishing industry and need for improving the technologies in the field of fishing industry.

Budget of States-UTs for development of Fisheries

Budget of States/UTs for development of Fisheries in State/UTs (Rs. In Crores)			
S. N	States/UTs	2019-20	2020-21
1	A & N Islands	29.00	31.00
2	Andhra Pradesh	564.95	335.75
7	Goa	88.25	86.55
13	Karnataka	320.00	291.62
14	Kerala	396.00	321.10
15	Lakshadweep	16.06	15.97
16	Madhya Pradesh	46.71	46.57
17	Maharashtra	260.00	182.00
22	Odisha	180.00	315.00
23	Pondicherry	60.90	26.92
27	Tamil Nadu	927.35	1229.85
28	Telangana	11.04	138.38
32	West Bengal	408.17	425.49
Source: Department of Fisheries, State Govt. / UTs Administration			

Share of Gross Value Addition(GVA) of Fisheries Sector (Constant Price: 2011-12)			
Years	GVA of fisheries Sector (Rs. in crore)	National GVA (Rs. in crore)	Share of Fisheries Sector (%)
2009-10	61,269	71,31,836	0.86
2010-11	64,663	77,04,514	0.84
2011-12	68,027	81,06,946	0.84
2012-13	71,362	85,46,275	0.84
2013-14	76,487	90,63,649	0.84
2014-15	82,232	97,12,133	0.85
2015-16	90,205	1,04,91,870	0.86
2016-17	99,627	1,13,28,285	0.88
2017-18	1,14,248	1,20,74,413	0.95
2018-19	1,28,011	1,28,03,128	1.00

Source: National Statistical Office, Ministry of Statistics and Programme Implementation, Government of India

This statistics gave an view that the boom of fishing industry and accounts to the major contribution for Indian Economy.

1.3 SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals (SDGs) include several targets related to fisheries and marine ecosystems.

1.Goal 14: Life Below Water - This goal aims to conserve and sustainably use the oceans, seas and marine resources. Its targets include:

- Target 14.4: By 2020, effectively regulate harvesting and end overfishing, illegal, unreported and unregulated (IUU) fishing and destructive fishing practices and implement science-based management plans, in order to restore fish stocks in the shortest time feasible, at least to levels that can produce maximum sustainable yield as determined by their biological characteristics.

- Target 14.6: By 2020, prohibit certain forms of fisheries subsidies which contribute to overcapacity and overfishing, eliminate subsidies that contribute to IUU fishing and refrain from introducing new such subsidies, recognizing that appropriate and effective special and differential treatment for developing and least developed countries should be an integral part of the World Trade Organization fisheries subsidies negotiation.

2.Goal 2: Zero Hunger - This goal aims to end hunger, achieve food security and improve nutrition, and promote sustainable agriculture. Its targets include:

- Target 2.3: By 2030, double the agricultural productivity and incomes of small-scale food producers, in particular women, indigenous peoples, family farmers, pastoralists and fishers, including through secure and equal access to land, other productive resources and inputs, knowledge, financial services, markets and opportunities for value addition and non-farm employment.

3.Goal 12: Responsible Consumption and Production - This goal aims to ensure sustainable consumption and production patterns. Its targets include:

- Target 12.2: By 2030, achieve the sustainable management and efficient use of natural resources.

- Target 12.5: By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse.

Overall, the SDGs recognize the importance of sustainable fisheries management for achieving food security, poverty reduction, and conservation of marine ecosystems. Achieving these goals will require collaboration among governments, fishing communities, scientists, and other stakeholders to implement science-based management plans and reduce the impacts of overfishing, IUU fishing, and destructive fishing practices.



From this statistical study through from the fishery department and SDG goals there is need for mapping marine fishing zones with the latest technologies having sustainable outcome.

1.4 INCOIS – Indian National Centre for Ocean Information Services.

Utilizing the remotely sensed data available from various satellites, the data on Sea Surface Temperature (SST) and Chlorophyll retrieved regularly from thermal-infrared channels of MODIS Aqua (USA) satellites are used for the identification of Potential Fishing Zones (PFZ) along the Indian coastline. Due to the dynamic nature of the ocean the fishing zones identified on the maps could shift from the marked location. This information helps the fishermen in locating the PFZ identified in the maps even if they reach the location after a day. The information is reaching to the coastal areas for more than a decade now. The feedback obtained from researchers working on various extensive validation projects as well as the feedback collected from fishermen indicated that the catch in the PFZ area is substantially higher when compared to the other areas. PFZ advisories were found more beneficial to artisanal, motorised and small mechanised sector fishermen engaged in pelagic fishing activities such as ring seining, gill netting etc., thereby reducing the searching time which in turn result in the saving of valuable fuel oil and also human effort.

1.5 PARAMETERS

The changes in the sea surface chlorophyll-a concentration (SSCC) and sea surface temperature (SST) are the most critical parameters that used for identifying fish availability as well as considered for estimating potential fishing zones (PFZ). Both parameters synergistically influence marine biota availability, ranging from plankton to marine mammals. The SST and other environmental factors (pH, oxygen, salinity) would covary depending on the region characteristics. For different environments variables, there are different characteristics for determining the PFZ.

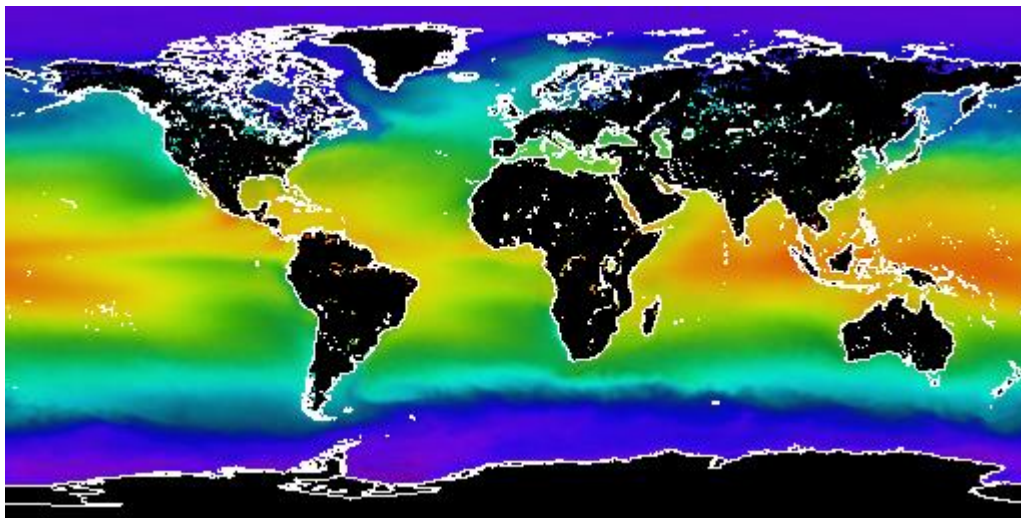
Additionally, SSCC is a proxy for phytoplankton biomass and is used to identify habitats of fish resources. The PFZ is a short-term prediction and credible on the zone of fishes aggregation in the open sea. It also explained that the PFZ is the zone where the fishes aggregate due to a food abundance. Their availability is limited by this zone in the open sea, where a sharp SST gradient with optimal SSCC co-occurrence at a given period.. Appropriately estimating the PFZ can optimize the fishing operation schedule.

In recent decades, the utilization of remote sensing technology and geographic information systems (GIS) have been conducted to analyse SSCC and SST for estimating the PFZ as well as for identifying the distribution patterns of fish and their interactions with other factors.

MODIS algorithms have been of greater value for oceanographic data extraction (e.g., SST and SSCC data). Using in-situ data from INCOIS forecast helps test the PFZ validity .This present research identified PFZ's and analysing their sustainability using MODIS data for September, October, November, December & January.

1.6 Sea Surface Temperature (SST)

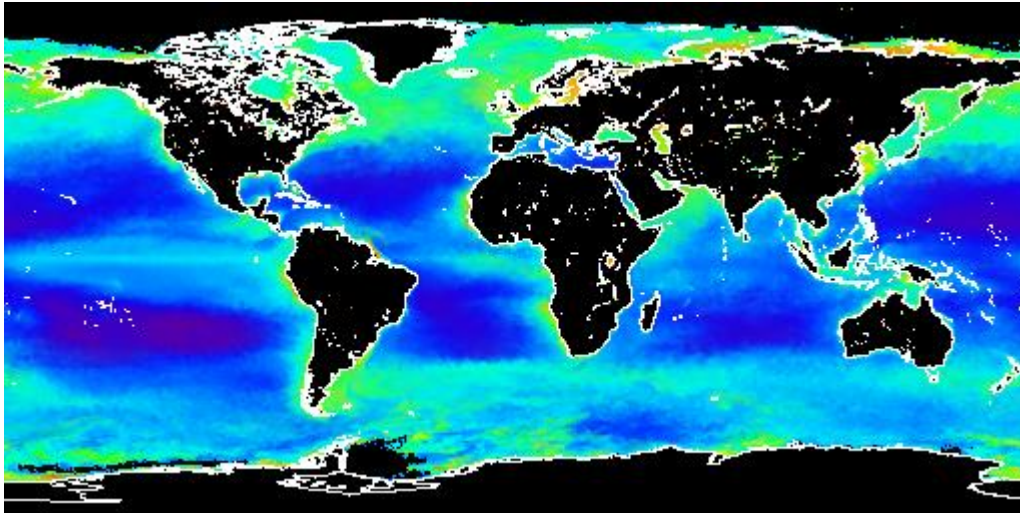
The hotness of the water near to the ocean's veneer is studied as Sea Surface Temperature (SST) or Ocean Surface Temperature (OST). Surface level differing is controlled by the measuring technique, yet it is a link 1 millimetre (0.04 in) to 20 metres (70 ft) less than sea surface. The environment in which fish live depends heavily on oceanographic conditions in these waters. Therefore, knowledge of conditions and changes in oceanographic factors needed to determine the exact fishing areas. One of the oceanographic factors that greatly affect the distribution and abundance of fish is the temperature of the waters.



WORLD Sea Surface Temperature (11 μ) COVER - [NASA Ocean Color](#)

1.7 CHLOROPHYLL-A

The distribution of chlorophyll-a density at the sea varied geographically as well as by the depth of the water. That variation was caused by the difference of sunlight intensity, and the concentration of nutrients within the waters. At sea, distribution chlorophyll-a density had a higher concentration at the beach and coastal areas and lowered at the offshore. The high distribution of chlorophyll-a concentrations in coastal waters is due to the supply of nutrients in large quantities through runoff from the mainland, whereas the low concentration of chlorophyll-a in the offshore area was because of the absence of direct supply of nutrients from the land. However, in certain of offshore areas there was a considerable amount of chlorophyll-a concentration. The situation is caused by the high concentration of nutrients produced through the physical process of water mass, where the water mass in lifting nutrients from the inner layer to the surface layer.



WORLD Chlorophyll-a COVER - [NASA Ocean Color](https://oceancolor.gsfc.nasa.gov/)

1.8 PROBLEM STATEMENT

1. The search for **fish frequently results in significant time and resource expenditure**, which **raises the cost and reduces profitability**.
2. The **fishing community will benefit from a reliable and timely advice** on the prospective fish aggregation zones because it will cut down on the time and effort spent searching for fish shoals, increasing profitability and, ultimately, socioeconomic standing.

1.9 OBJECTIVES

1. Application of **satellite-derived sea surface temperature (SST) and chlorophyll a** together in mapping marine fishing zones .
2. Developing an **index for finding most favorable location** for fishing.

CHAPTER 2

2.1 LITERATURE REVIEW

Mapping marine fisheries using geoinformatics is a rapidly growing field of research and is becoming increasingly important in the sustainable management of marine resources. Geoinformatics refers to the use of geographical information systems (GIS), remote sensing, and other spatial technologies to study and analyse the natural environment. In the case of marine fisheries, these tools can be used to map fishing grounds, track fishing vessels, and monitor fish populations, among other things.

1. A study by **David et al. (2017)** highlights the importance of geoinformatics in mapping marine fisheries in the Indian Ocean. The study used satellite-based remote sensing data to identify fishing hotspots, where there was a high concentration of fishing activity. This information was then used to develop a model that could predict the location of fishing activity based on factors such as sea surface temperature, chlorophyll-a concentration, and sea surface height.
2. Similarly, a study by **Suryanarayana et al. (2016)** used GIS and remote sensing to map the spatial distribution of the tuna fishery in the Indian Ocean. The study found that the tuna fishery was concentrated in a few key areas, and that these areas shifted seasonally in response to changes in ocean conditions. This information can be used to better manage the tuna fishery and ensure its sustainability.
3. In the Pacific Ocean, a study by Hulbert et al. (2020) used acoustic surveys and GIS to map the distribution of fish populations in the coral reef ecosystem. The study found that fish populations were higher in areas with more coral cover, and that these areas were also more resilient to disturbances such as cyclones and coral bleaching. This information can be used to develop strategies to protect coral reefs and the fish populations that depend on them.
4. In addition to mapping fishing grounds and fish populations, geoinformatics can also be used to track fishing vessels and monitor their activities. A study by Tsou et al. (2016) used satellite data to track the movement of fishing vessels in the South China Sea. The study found that vessels tended to concentrate in certain areas, and that this concentration was linked to the presence of fish stocks. This information can be used to develop strategies to manage fishing activities and prevent overfishing.
5. Overall, the use of geoinformatics for mapping marine fisheries is a promising field of research that has the potential to inform more sustainable management practices. By mapping fishing grounds, tracking fishing vessels, and monitoring fish populations, researchers and managers can make more informed decisions about how to protect and manage marine resources for the benefit of both humans and the environment.

Here are some papers related to using geoinformatics for mapping marine fisheries:

1. David, G., Sathyendranath, S., & Platt, T. (2017). Using remote sensing to map the spatial distribution of fishing effort in the Indian Ocean. *Remote Sensing of Environment*, 190, 305-318.

2. Suryanarayana, P., Nayak, S. R., Chakraborty, S., & Nayak, B. B. (2016). Geospatial analysis of the spatial distribution of tuna fishery in the Indian Ocean. *Marine Policy*, 64, 54-63.
3. Hulbert, J. M., McWilliam, J. N., Williams, I. D., & Friedlander, A. M. (2020). Mapping the spatial distribution of fish communities and habitats in the coral reef ecosystem of the Pacific Remote Island Areas. *PloS one*, 15(1), e0227556.
4. Tsou, M. H., Lin, Y. T., Huang, C. J., & Lin, C. L. (2016). Identifying fishing ground hotspots in the South China Sea using satellite remote sensing and geographic information systems. *Environmental Monitoring and Assessment*, 188(11), 629.
5. Kleisner, K. M., Longo, C., Halpern, B. S., Hardy, D., Best, B. D., Frazier, M., ... & Katona, S. K. (2016). Mapping fishing effort through AIS data: Initial assessment of the challenges and uncertainties of using big data methods to map fishing effort. *PloS one*, 11(5), e0155760.
6. Batsukh, N., Hong, S. Y., Park, K. J., Kim, T. I., & Seo, H. (2019). A GIS-based mapping and analysis of fishing grounds around the Korean Peninsula. *Ocean & Coastal Management*, 180, 104938.

These papers demonstrate the wide range of applications for geoinformatics in mapping marine fisheries, from identifying fishing hotspots to tracking fishing vessels and mapping fish populations.

1. Ramachandran, C., & Vidyasagar, K. (2018). Indian marine fisheries—challenges and opportunities. *Reviews in Fish Biology and Fisheries*, 28(4), 669-691.
2. Thangavelu, R., Ramesh, R., & Narayanasamy, R. (2017). Analysis of marine fisheries resources and fisheries management in India. *Environmental Development*, 22, 12-22.
3. Srinivasan, M., & Dineshbabu, A. P. (2017). Marine fisheries in India: the challenges and the opportunities. *Current Science*, 112(2), 221-230.
4. Thirumilu, P., & Rajasekar, K. T. (2016). Spatial analysis of marine fishery resources along the coast of Tamil Nadu, India. *Environmental Monitoring and Assessment*, 188(8), 450.
5. James, P. S. B. R. (2016). Status of marine fisheries in India. *Indian Journal of Fisheries*, 63(1), 1-13.

These papers cover a range of topics related to Indian marine fisheries, including challenges and opportunities, fisheries management, and spatial analysis of fishery resources. They provide insights into the current state of Indian marine fisheries and offer suggestions for sustainable management and conservation.

CHAPTER 3

3.1 STUDY AREA

The Indian coastal zone stretches over 8,100 km, with a diverse range of habitats, from mangroves and estuaries to rocky shores and coral reefs. The Indian coast stretches from the Arabian Sea to the west, the Bay of Bengal to the east, and the Indian Ocean to the south.

The Indian coast is divided into nine states: Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Odisha, and West Bengal. These states have their own unique cultures, languages, and cuisine, and the coastal regions are often known for their seafood dishes. Some of the major ports along the Indian coast include Mumbai, Chennai, Kolkata, Kochi, and Visakhapatnam, among others. These ports are important hubs for trade, fishing, and tourism. The Indian coast is also home to several important marine protected areas, including the Gulf of Mannar Biosphere Reserve, the Great Nicobar Biosphere Reserve, and the Sundarbans Biosphere Reserve, among others. These protected areas are important for conserving marine biodiversity, as well as for supporting the livelihoods of local communities.

Overall, the Indian coast is a vital part of the country's culture, economy, and biodiversity, and is an important area of research and conservation.

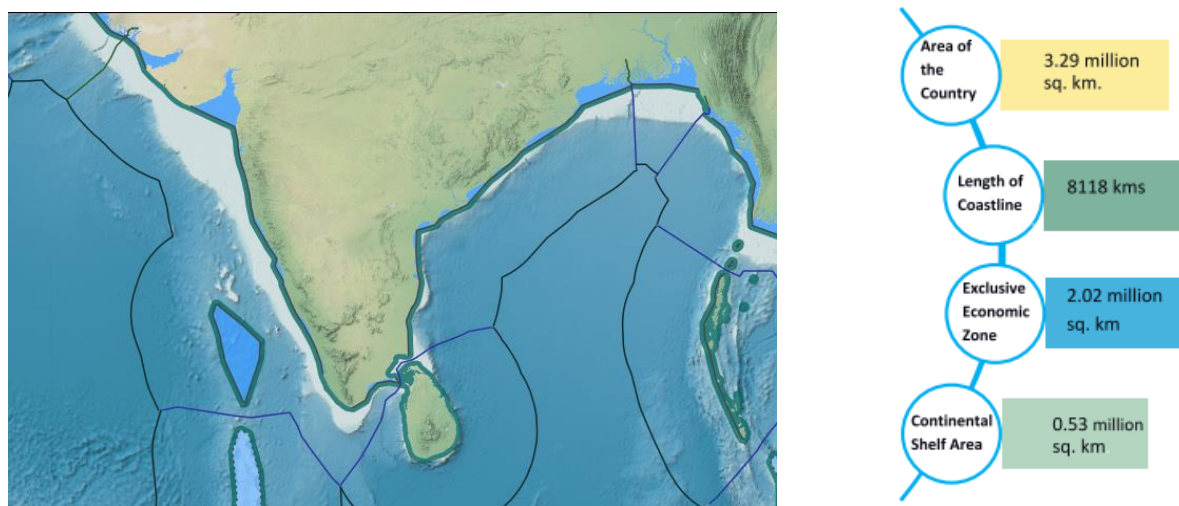


FIG. INDIAN COAST

CHAPTER 4

4.1 DATA SOURCES

Table : Data Sources

PARAMETERS	SENSORS	RESOLUTION
Sea Surface Temperature (SST)	MODIS-OCM	4.6 Km
Chlorophyll-a	MODIS-OCM	4.6 Km

4.2 Data availability

1. Sea Surface Temperature: [Ocean Color Browse : Specify Search Parameters \(nasa.gov\)](#)
2. Chlorophyll-a: [Ocean Color Browse : Specify Search Parameters \(nasa.gov\)](#)

4.3 MODIS- OCM

The Moderate Resolution Imaging Spectroradiometer (MODIS) is a primary instrument on the Terra (formerly EOS AM-1) and Aqua (formerly EOS PM-1) satellites. Terra's orbit around the Earth is scheduled such that it crosses the equator from north to south in the morning, whereas Aqua crosses the equator from south to north in the afternoon. Every 1 to 2 days, Terra MODIS and Aqua MODIS observe the whole Earth's surface, collecting data in 36 spectral bands, or sets of wavelengths (see MODIS Technical Specifications). These findings will help us better comprehend global dynamics and processes on land, in the seas, and in the lower atmosphere. MODIS is a key player in the creation of verified, global, interactive Earth system models that can reliably anticipate global change and help policymakers make informed decisions about environmental protection.

CHAPTER 5

5.1 METHODOLOGY

CRITERIA EMPLOYED IN

From the literature review done the following are the grades of potential fishing zone and the conditions of parameters range.

No	PFZ Category	Temperature Criteria (°C)	Chlorophyll-a Criteria (mg/m ³)
1	Potential	24 - 27	> 0.2
2	Medium potential	27 - 30	0.1 – 0.2
3	Less potential	<24 - >30	< 0.1

Although this a known criteria, Indian ocean has its own range as tabulated below.

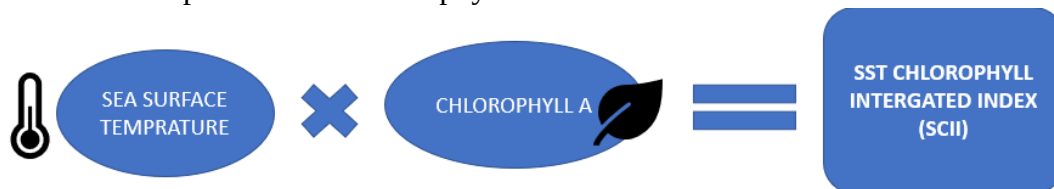
TABLE: SST and CHLOR_A variation for the study area.

SST (°C)	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER	JANUARY
MIN	24.17	28.22	27.68	25.87	24.17
MAX	28.7	29.57	29.82	29.51	28.7

CHL-a(mg/m ³)	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER	JANUARY
MIN	0.052	0.062	0.05664	0.079	0.077
MAX	3.684	3.8	2.56	3.161	2.47

5.2 SCII- SST Chlorophyll Integrated Index

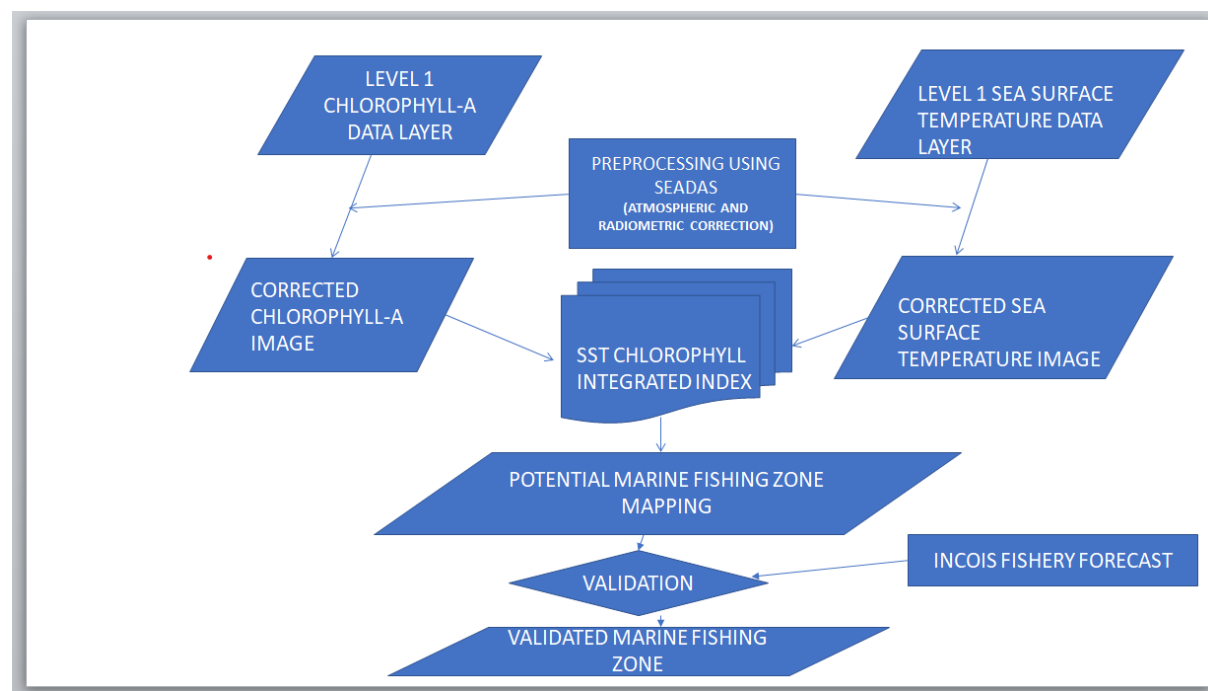
Considering the factors typically for the Indian coast, as the temperature variation for few months are low, grading of PFZ is having low separability. Hence, a new index was developed termed as **SCII – SST Chlorophyll Integrated Index** by getting the product of Sea Surface Temperature and Chlorophyll a .



This is specifically for the Indian coast dynamics.

SCII	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER	JANUARY
MIN	1.45	1.79	1.64	2.264	2.13
MAX	95.81	109.21	72.35	82.89	63.63

5.3 METHODOLOGY FLOW CHART



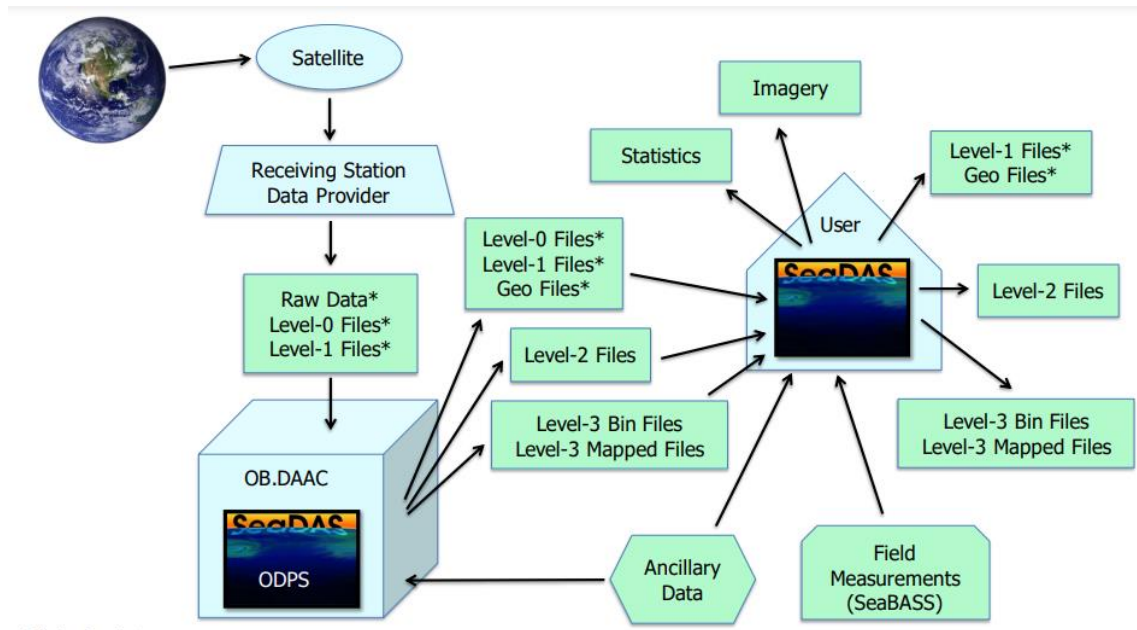
Level-1 Chlorophyll-a and level-1 SST data were downloaded from NASA OCEAN COLOUR. Pre-Processing was done using SEADAS software, which is an open source software provided by NASA. Resampling, subsetting, atmospheric correction were done using SEADAS. Thus, we got corrected Chlorophyll-a and SST images. A union operation was done between these layers to get Potential Fishing Zones. The criteria of Chlorophyll-a is $> 0.2\text{mg/m}^3$ and SST is between 24°C and 27°C . A map of this union is Potential Fishing Zones map. To validate our work, we took INCOIS fishery forecast data. And hence, a validated Potential Fishing Zones map was obtained.

5.4 SOFTWARE

SeaDAS is developed by NASA. SeaDAS is a comprehensive software package for the processing, display, analysis, and quality control of remote-sensing Earth data. SeaDAS currently supports the science processing of over 15 US and international satellite missions. Additional coinciding ancillary data are used to correct for and calibrate out the atmospheric components of the signal in order to determine an Earth/ocean surface level signal.



PREPROCESSING



As for Pre-Processing, Level-1 data was downloaded.

Data pre-processing required a client-server architecture as the data was not able to process in a Windows OS. A server called Docker was used to access a Linux OS.

As the data was huge and incompatible, pre-processing was not completely fulfilled.

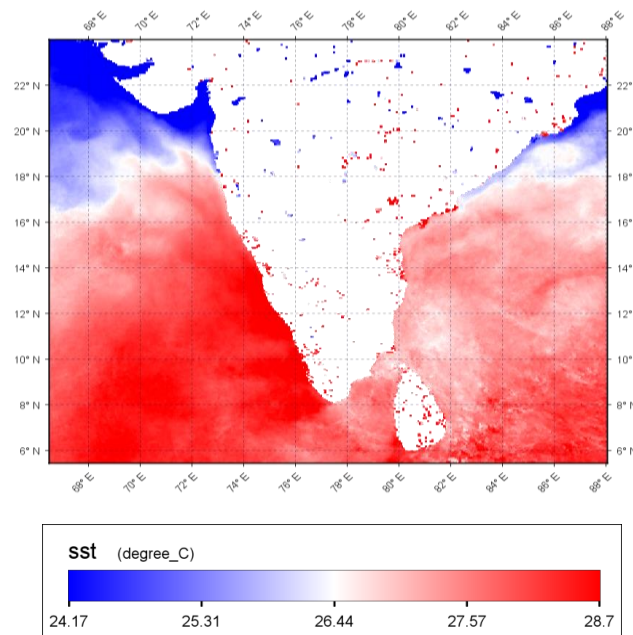
Later level-3 data was downloaded from NASA OCEAN COLOUR website.

CHAPTER 6

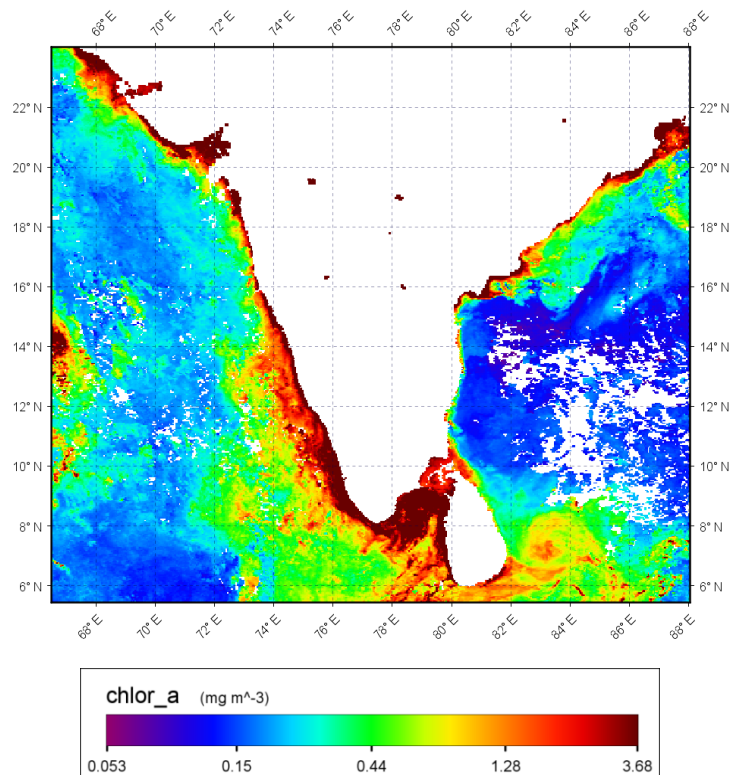
6.1 RESULTS AND DISCUSSION

SEPTEMBER

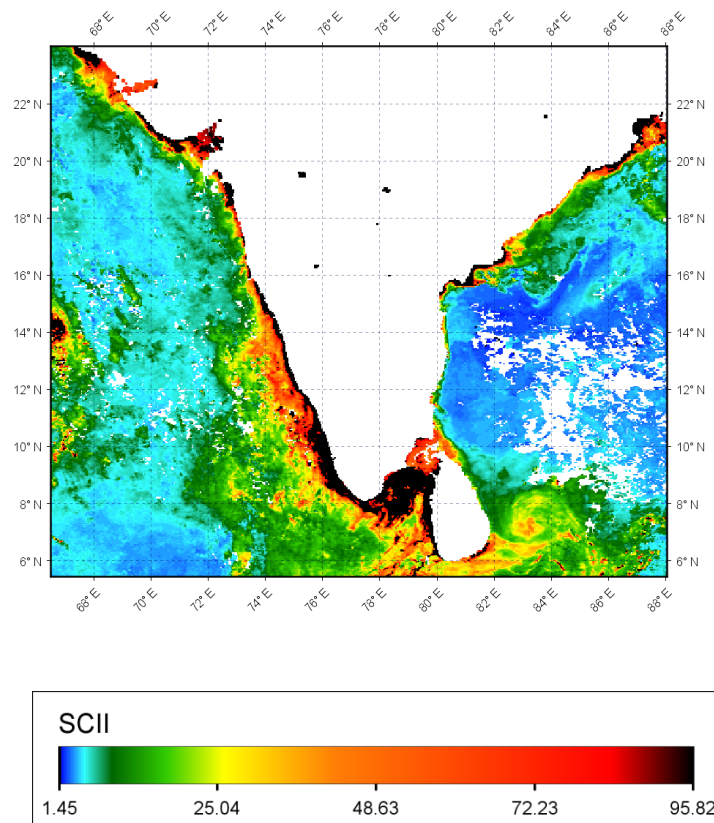
1. SEA SURFACE TEMPERATURE



2. CHLOROPHYLL-a



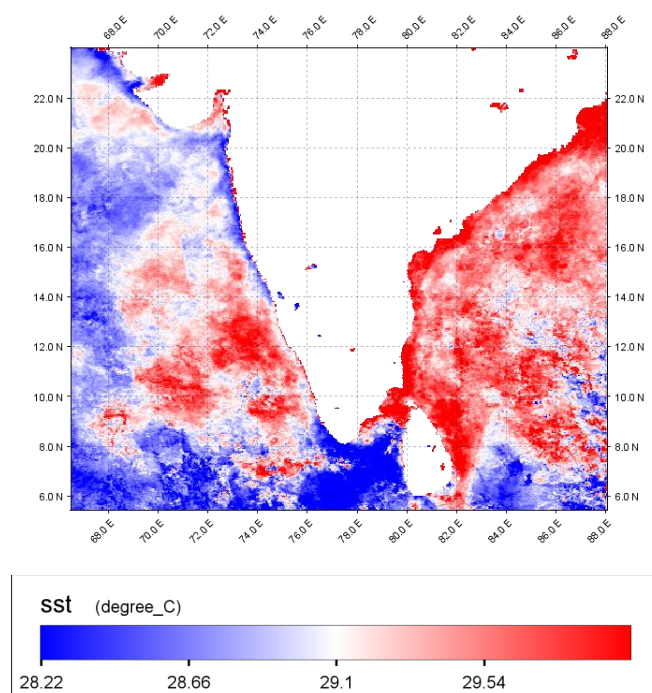
3. SST-CHLOROPHYLL INTEGRATED INDEX



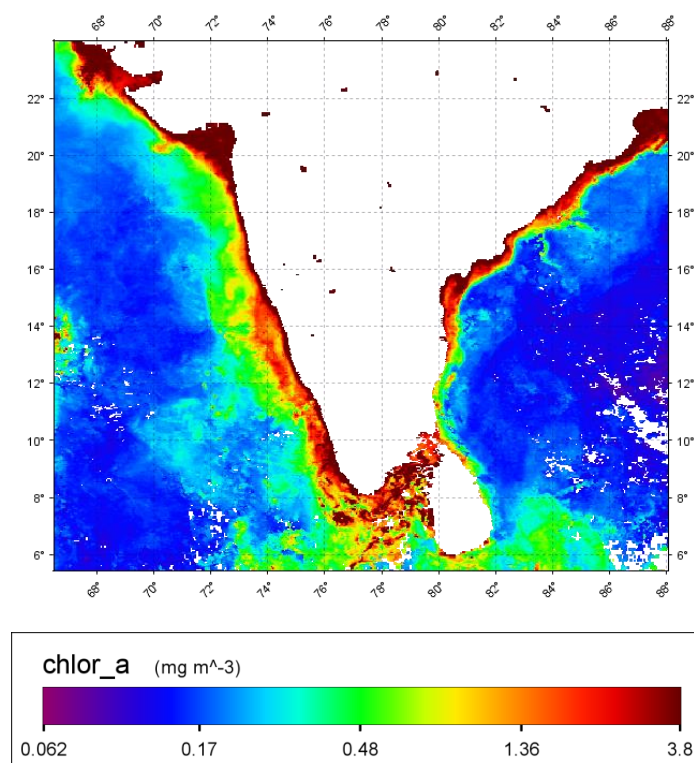
1. In the month of September, SST varied from 24°C to 28°C. Most part of Indian Ocean had a temperature of more than 27°C, whereas northern part of the Indian Ocean was on a cooler side.
2. There was good spread of chlorophyll content around Indian coast. Cyclone Sitarang had a role in this spread over Bay of Bengal.
3. Western part of the southernmost tip of India can be seen as a potential fishing zone. Also around Andaman and Nicobar islands showed a good PFZ.

OCTOBER

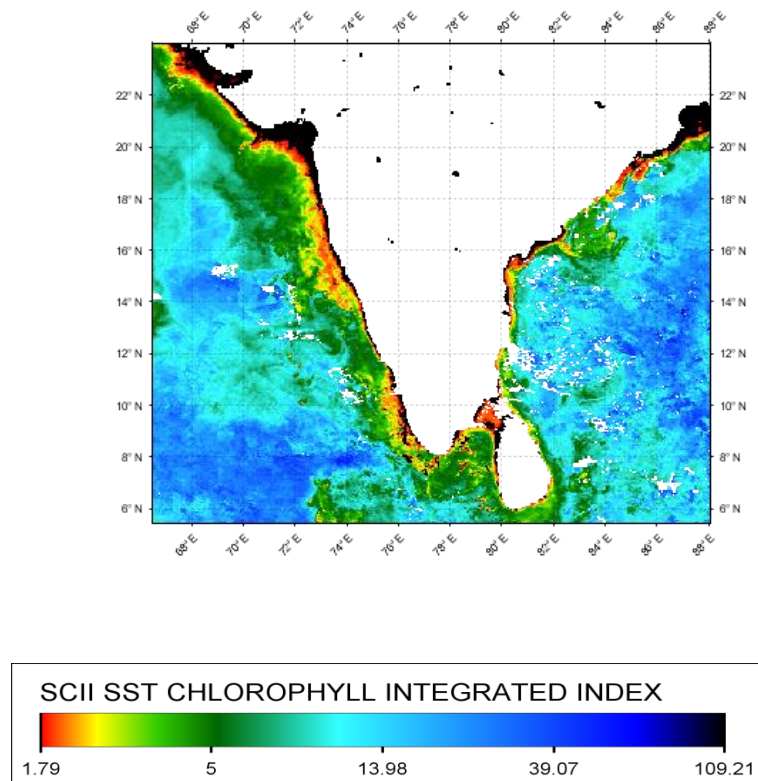
1. SEA SURFACE TEMPERATURE



2. CHLOROPHYLL-a



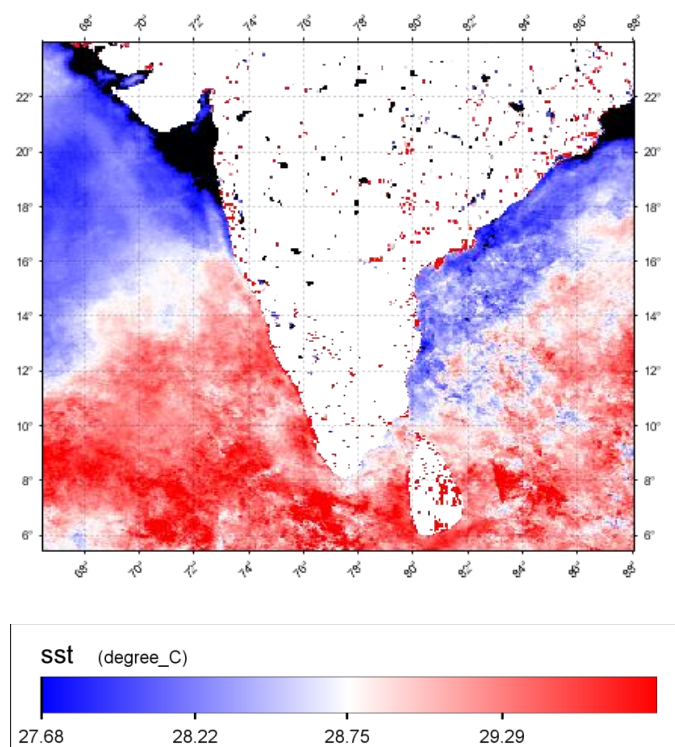
3. SST-CHLOROPHYLL INTEGRATED INDEX



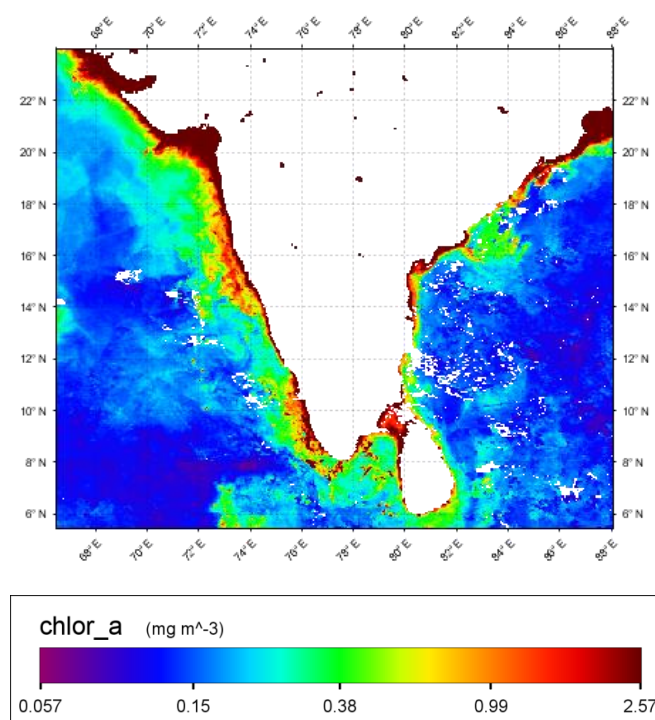
1. In the month of October, SST varied from 26°C to 31°C. Bay of Bengal monthly average temperature was on a higher side of 29°C.
2. Chlorophyll-a showed a plain pattern. Few kilometers away from the coast showed adequate chlorophyll value for fishes.
3. Index showed the wide spread of probable fishing zones. Fishermen had to go afar from coast for good quantity of fish.

NOVEMBER

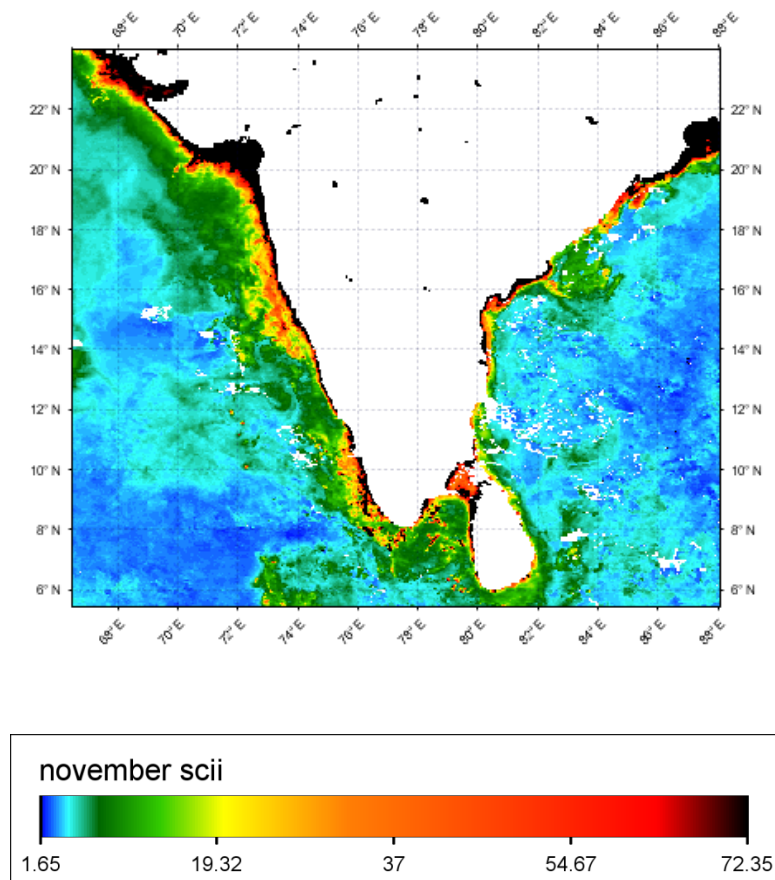
1. SEA SUREFACE TEMPERATURE



2. CHLOROPHYLL-a



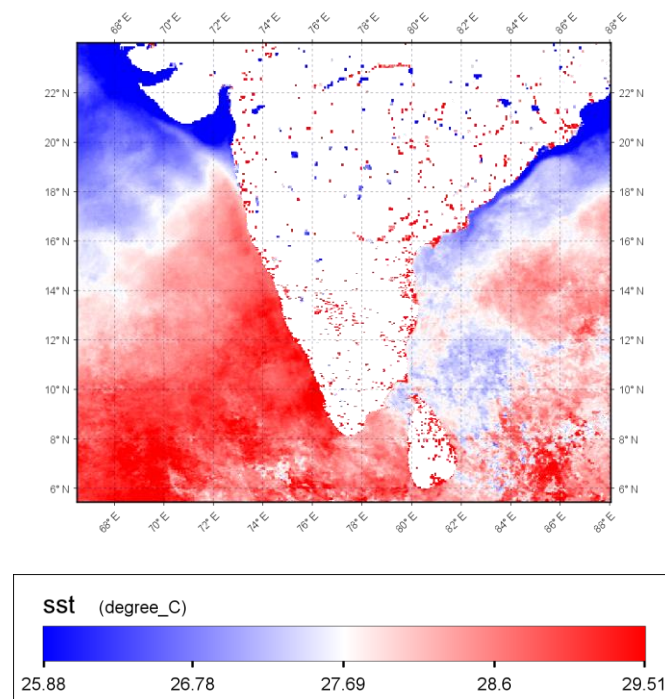
3. SEA SURFACE TEMPERATURE CHLOROPHYLL INDEX



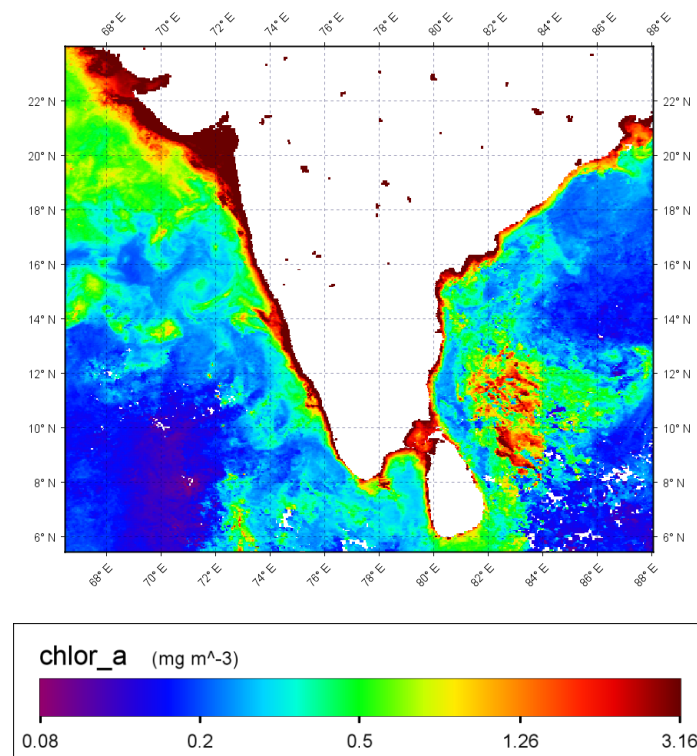
1. In the month of November, SST varied from 27°C to 30°C. Coast of Orissa and West Bengal had a cooler temperature in November when compared to other months.
2. Chlorophyll-a spread was similar to the month of October.
3. Index also showed similar results to that of month of October.

DECEMBER

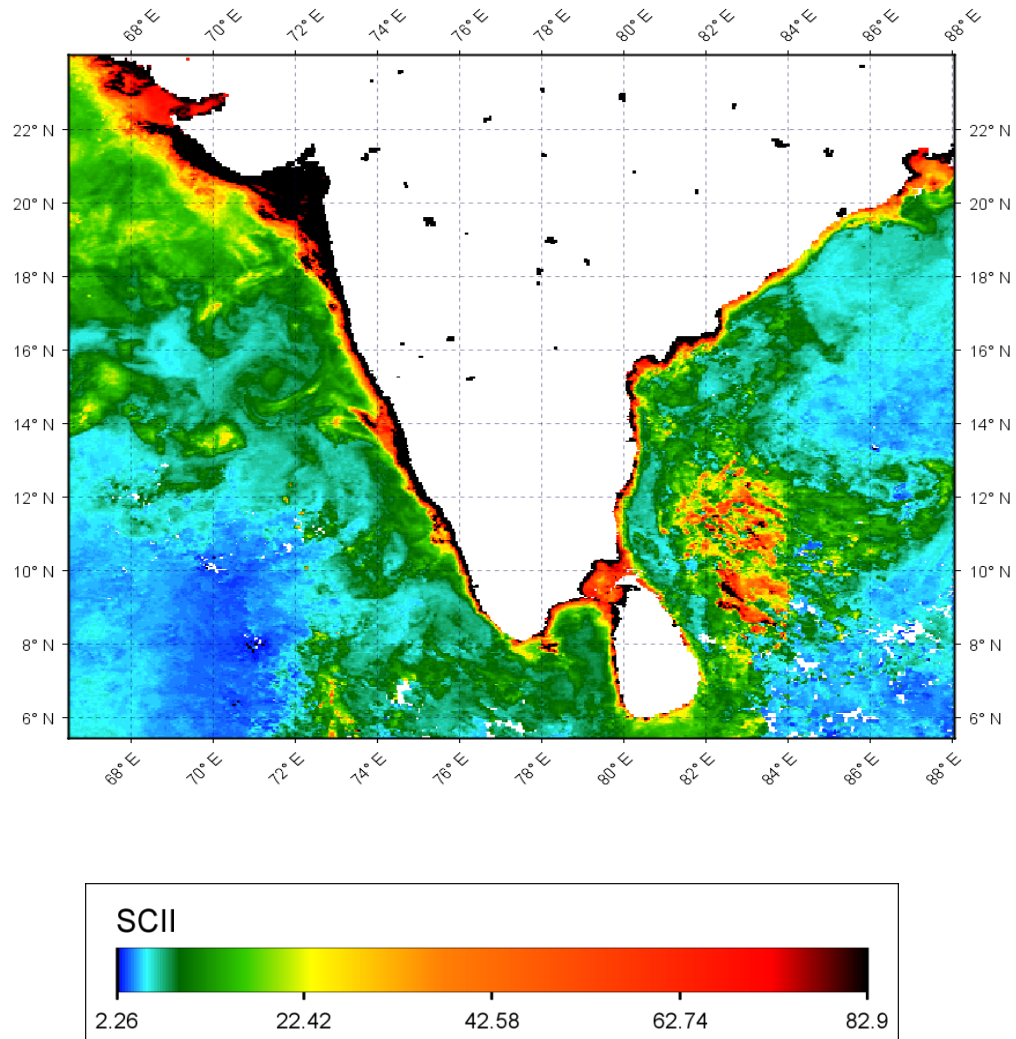
1. SEA SURFACE TEMPERATURE



2. CHLOROPHYLL-a



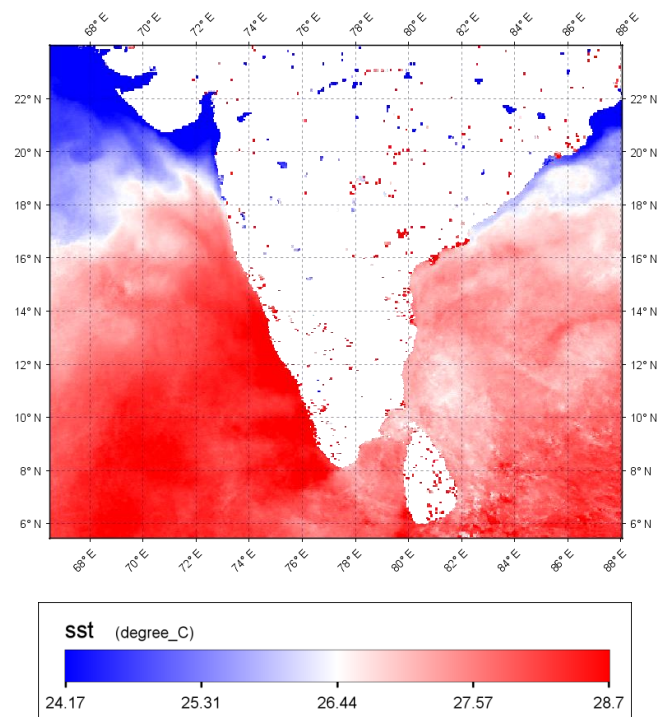
3. SEA SURFACE TEMPERATURE CHLOROPHYLL INDEX



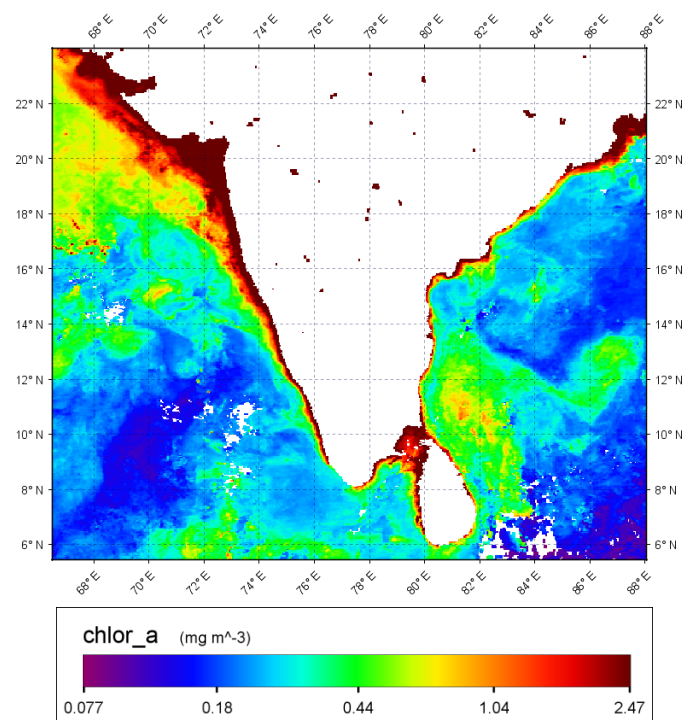
1. In the month of December, northern part of Indian Ocean and coastal regions of Orissa and Tamil Nadu had cooler temperature.
2. Coastal part of Gujarat and Southern part of Bay of Bengal has higher quantities of Chlorophyll-a.
3. Month of December showed more PFZ's compared to other months than January. Few kilometers away from coast would give the fishermen a good catch.

JANUARY

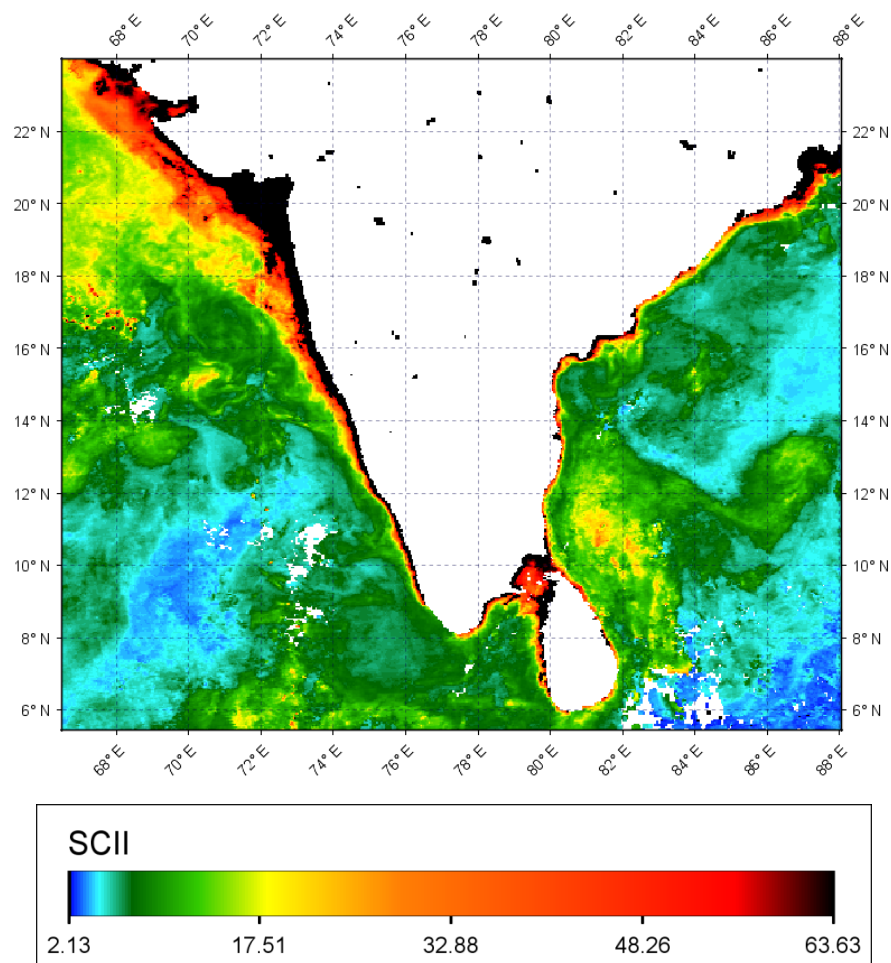
1. SEA SURFACE TEMPERATURE



2. CHLOROPHYLL-a

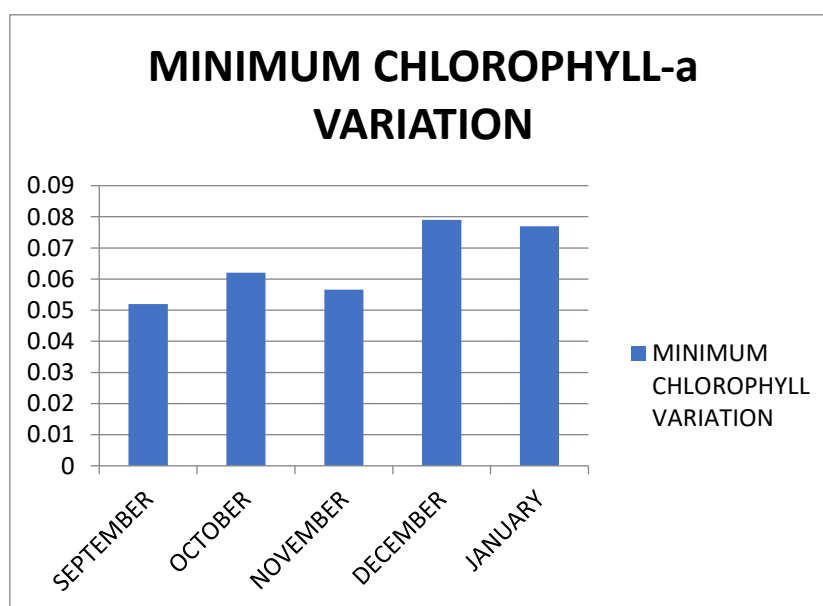
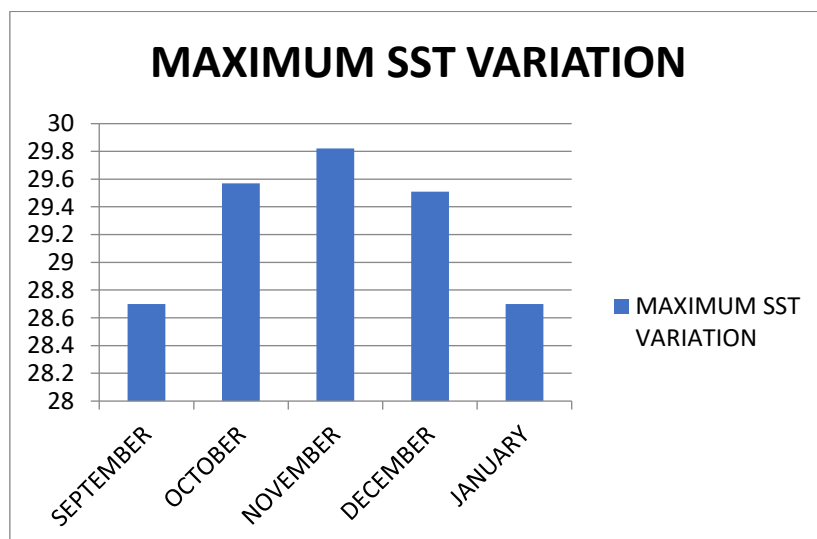
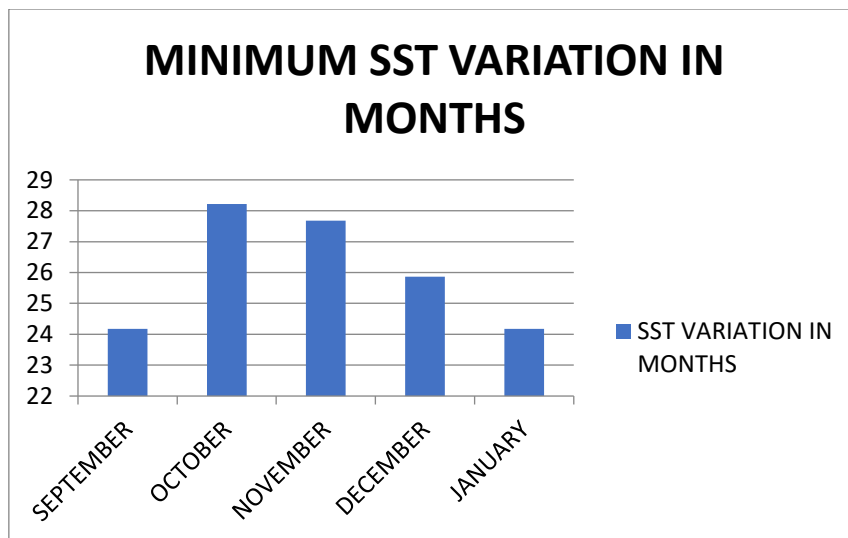


3. SEA SURFACE TEMPERATURE –CHLOROPHYLL INTEGRATED INDEX

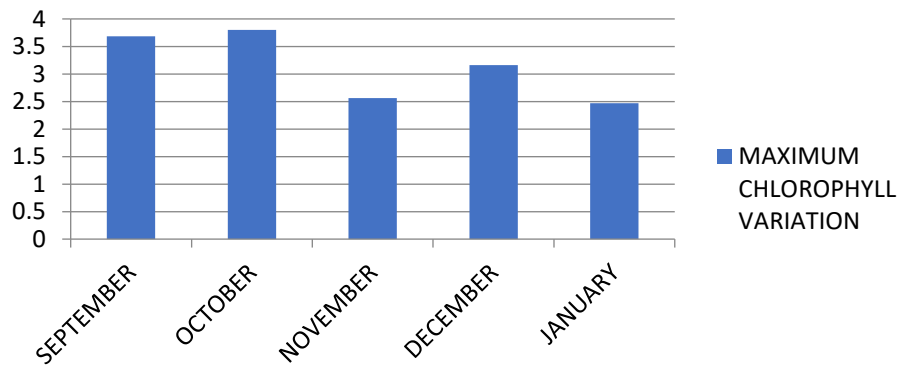


1. In the month of January, SST was ideal for PFZ which is 24°C to 27°C. South-west coast SST was on a higher side
2. There was also a good spread of chlorophyll-a content around Indian Ocean.
3. January was the most ideal month for fishes. Both SST and chlorophyll-a parameters were adequate.

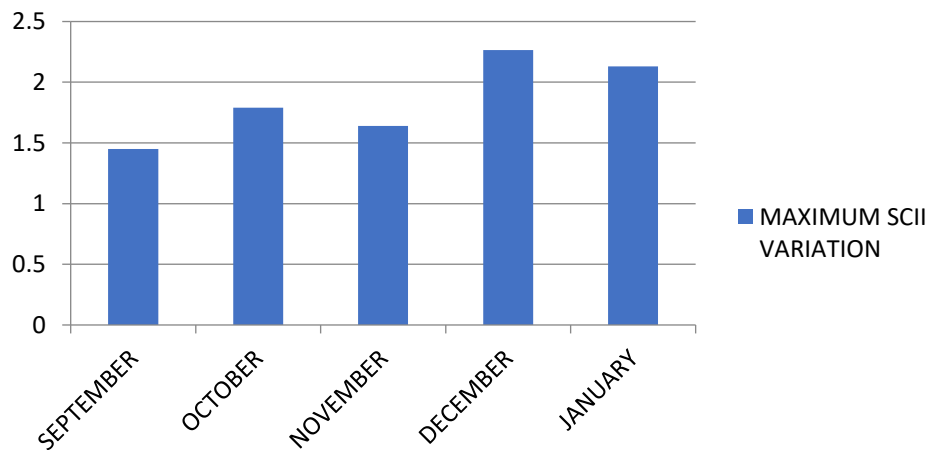
6.2 GRAPHS SHOWING VARIATIONS



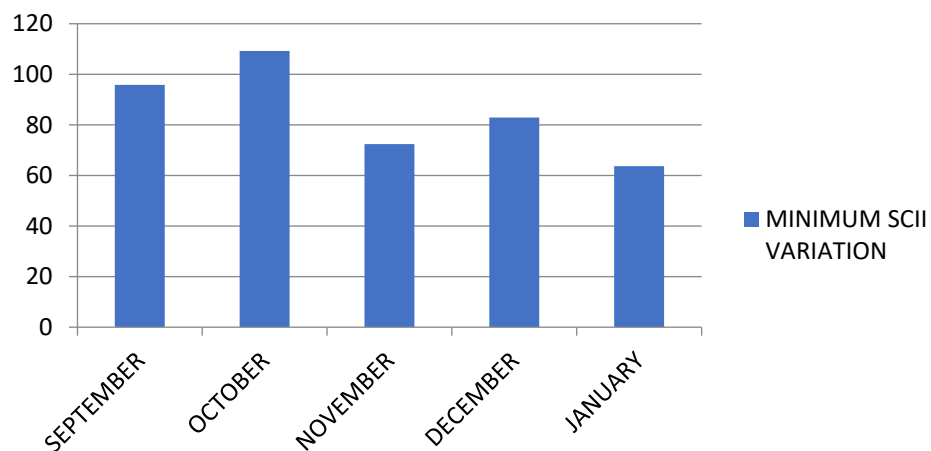
MAXIMUM CHLOROPHYLL-a VARIATION



MINIMUM SCII VARIATION



MAXIMUM SCII VARIATION



6.3 SEA TRUTH

KARNATAKA FORECAST VALIDITY FROM 15 JAN 2023 TO 16 JAN 2023						
From the coast of	Direction	Bearing (deg)	Distance (km) From-To	Depth (mtr) From-To	Latitude (dms)	Longitude (dms)
Shirali	SW	235	20-25	32-37	13 54 57 N	74 20 35 E
Kasarkod	SW	253	32-37	55-60	14 9 35 N	74 8 6 E
Mavalli	NW	278	27-32	37-42	14 8 26 N	74 13 18 E
Gangavali	SW	234	37-42	47-52	14 22 43 N	74 0 1 E
Bhatkal	SW	241	19-24	34-39	13 52 53 N	74 21 32 E
Hangarkatta	NW	298	39-44	36-41	13 36 57 N	74 22 6 E
Mulki	NW	319	56-61	35-40	13 28 59 N	74 25 33 E
Hosabettu-Udaivar	NW	321	56-61	28-33	13 27 0 N	74 26 35 E
Dhadeshvar	SW	251	40-45	48-53	14 15 6 N	74 2 19 E
Navunda	NW	285	25-30	30-35	13 48 56 N	74 23 35 E
Kapu	NW	314	45-50	35-40	13 31 12 N	74 25 11 E
Belekeri	SW	222	40-45	48-53	14 25 23 N	73 59 50 E
Karki	SW	254	37-42	46-51	14 12 55 N	74 3 54 E
Kota	NW	288	38-43	37-42	13 37 21 N	74 19 56 E
Navayatkere	SW	260	30-35	43-48	14 8 39 N	74 10 38 E
Udiyavara	NW	309	42-47	34-39	13 33 8 N	74 24 6 E
Coondapoor (gangoli)	SW	267	37-42	40-45	13 36 4 N	74 18 21 E
Tadri	SW	241	39-44	41-46	14 20 2 N	74 0 13 E
Honavar	SW	254	36-41	45-50	14 11 3 N	74 5 51 E
Gangoli	NW	294	30-35	30-35	13 46 37 N	74 23 24 E
Kumta Pt	SW	249	40-45	48-53	14 17 29 N	74 1 5 E
Baindur	SW	266	22-27	32-37	13 51 11 N	74 23 9 E
Malpe	NW	310	39-44	34-39	13 35 7 N	74 23 14 E

ABBREVIATION:

SE: South East

SW: South West

NE: North East

NW: North West

N: North

E: East

deg: Degree

km: Kilometres

mi: Miles

nmi: Nautical Miles

mtr: Meters

fth: Fathom

dd: Degree Decimal

dms: Degree Minute Second

dmm: Degrees Minute.m

The above table shows the INCOIS PFZ for the month of January. This data was used as sea truth data. Those INCOIS points matching with our data were given as 1 and the rest, which didn't match our data were given 0.

18 out of 25 points matched our data, showing 72% agreement.

CHAPTER 7

7.1 CONCLUSION

The current study focused on mapping marine fisheries for Indian Coast region considering the parameters of Sea Surface Temperature and Chlorophyll a as major for the months of September 2022- January 2023. The results were compared with INCOIS data and was found with 72% agreement. This factor gave a clue to consider additional factors of bathymetry and ocean movements in fishery zone prediction. The nature of Indian ocean is unique and dynamic a better understanding of this context is required to obtain better outcomes in mapping marine fisheries. This studied enhanced our study of dealing with ocean parameters , INCOIS data interpretation and SEADAS software analysis.

CHAPTER 8

8.1 REFERENCES

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